Problem Statement: Write a simple program in SCALA using Apache Spark framework

Code:

Install Java:

```
vboxuser@Ubuntu:~$ sudo apt install openjdk-11-jdk
[sudo] password for vboxuser:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openjdk-11-jdk is already the newest version (11.0.26+4-1ubuntu1~24.04).
The following package was automatically installed and is no longer required:
  gnupg-l10n
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 122 not upgraded.
```

Install Scala:

```
vboxuser@Ubuntu:~$ sudo apt install scala
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  gnupg-l10n
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  libhawtjni-runtime-java libjansi-native-java libjansi1-java libjline2-java
  scala-library scala-parser-combinators scala-xml
Suggested packages:
  scala-doc
The following NEW packages will be installed:
  libhawtjni-runtime-java libjansi-native-java libjansi1-java libjline2-java
  scala scala-library scala-parser-combinators scala-xml
0 upgraded, 8 newly installed, 0 to remove and 122 not upgraded.
Need to get 25.1 MB of archives.
After this operation, 28.6 MB of additional disk space will be used.
Do you want to continue? [Y/n] Y
Get:1 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 libhawtjni-runtime-java all 1.18-1 [28.9 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 libjansi-native-java all 1.8-2 [24.1 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 libjansi1-java all 1.18-3ubuntu1 [56.0 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 libjline2-java all 2.14.6-5 [150 kB]
Get:5 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 scala-library all 2.11.12-5 [9,586 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 scala-parser-combinators all 1.0.3-3.1 [365 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 scala-xml all 1.0.3-3.1 [615 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 scala all 2.11.12-5 [14.2 MB]
Fetched 25.1 MB in 11s (2,339 kB/s)
Selecting previously unselected package libhawtjni-runtime-java.
(Reading database ... 188419 files and directories currently installed.)
```

```
Preparing to unpack .../0-libhawtjni-runtime-java_1.18-1_all.deb ...
Unpacking libhawtjni-runtime-java (1.18-1) ...
Selecting previously unselected package libjansi-native-java.
Preparing to unpack .../1-libjansi-native-java_1.8-2_all.deb ...
Unpacking libjansi-native-java (1.8-2) ...
Selecting previously unselected package libjansi1-java.
Preparing to unpack .../2-libjansi1-java_1.18-3ubuntu1_all.deb ...
Unpacking libjansi1-java (1.18-3ubuntu1) ...
Selecting previously unselected package libjline2-java.
Preparing to unpack .../3-libjline2-java_2.14.6-5_all.deb ...
Unpacking libjline2-java (2.14.6-5) ...
Selecting previously unselected package scala-library.
Preparing to unpack .../4-scala-library_2.11.12-5_all.deb ...
Unpacking scala-library (2.11.12-5) ...
Selecting previously unselected package scala-parser-combinators.
Preparing to unpack .../5-scala-parser-combinators_1.0.3-3.1_all.deb ...
Unpacking scala-parser-combinators (1.0.3-3.1) ...
Selecting previously unselected package scala-xml.
Preparing to unpack .../6-scala-xml_1.0.3-3.1_all.deb ...
Unpacking scala-xml (1.0.3-3.1) ...
Selecting previously unselected package scala.
Preparing to unpack .../7-scala_2.11.12-5_all.deb ...
Unpacking scala (2.11.12-5) ...
Setting up scala-library (2.11.12-5) ...
Setting up scala-xml (1.0.3-3.1) ...
Setting up scala-parser-combinators (1.0.3-3.1) ...
Setting up libhawtjni-runtime-java (1.18-1) ...
Setting up libjansi-native-java (1.8-2) ...
Setting up libjansi1-java (1.18-3ubuntu1) ...
Setting up libjline2-java (2.14.6-5) ...
Setting up scala (2.11.12-5) ...
update-alternatives: using /usr/share/scala-2.11/bin/scala to provide /usr/bin/s
```

Create a .scala file and write the code:

Open

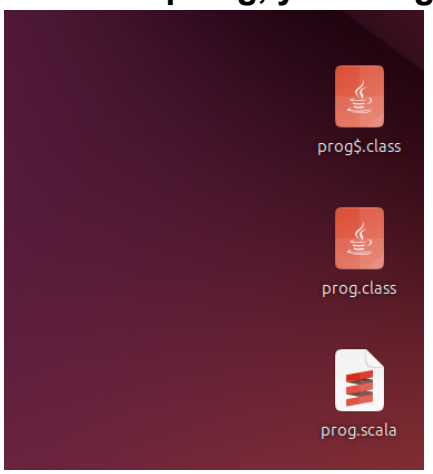
prog.scala
~/Desktop

```scala
object prog{
        def main(args: Array[String])
        {
                println("Hello World!")
        }
}
```

Compile the prog.scala file in terminal:

```
vboxuser@Ubuntu:~/Desktop$ scalac prog.scala
```

After compiling, you will get prog.class as an output in the same directory:



Run the program using following command:

```
vboxuser@Ubuntu:~/Desktop$ scala prog
Hello World!
vboxuser@Ubuntu:~/Desktop$
```